

Self-Supervised Representation Learning for UAV-Based Semantic Segmentation

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Motivation

Problem

UAVs provide ultra-high-resolution, low-cost imagery for urban scene analysis, but achieving accurate **semantic segmentation** remains **challenging**.

Limitation of existing methods:

- ❖ **Pixel-level annotations** are **expensive** and **limited**
- ❖ **ImageNet** features transfer **poorly** due to aerial viewpoints, scale variation, and small objects

Contributions

- ❑ A self-supervised pretraining approach that boosts UAV semantic segmentation performance without requiring any additional labeled data
- ❑ Finetuned across three architectures – UNet, UNet++, and DeepLabV3
- ❑ Consistent improvements in mIoU and rare-class IoU, showing domain-specific SSL outperforms ImageNet initialization

Datasets

❑ Unlabeled Datasets (SSL Pretraining)

Dataset	# Images	Resolution
UAVid	270	3840×2610
Semantic Drone	400	6000×4000
UDD	288	4000×3000

❑ Labeled Dataset (Supervised Finetuning)

- Train: 200 images
- Validation: 70 images
- Test: 150 images

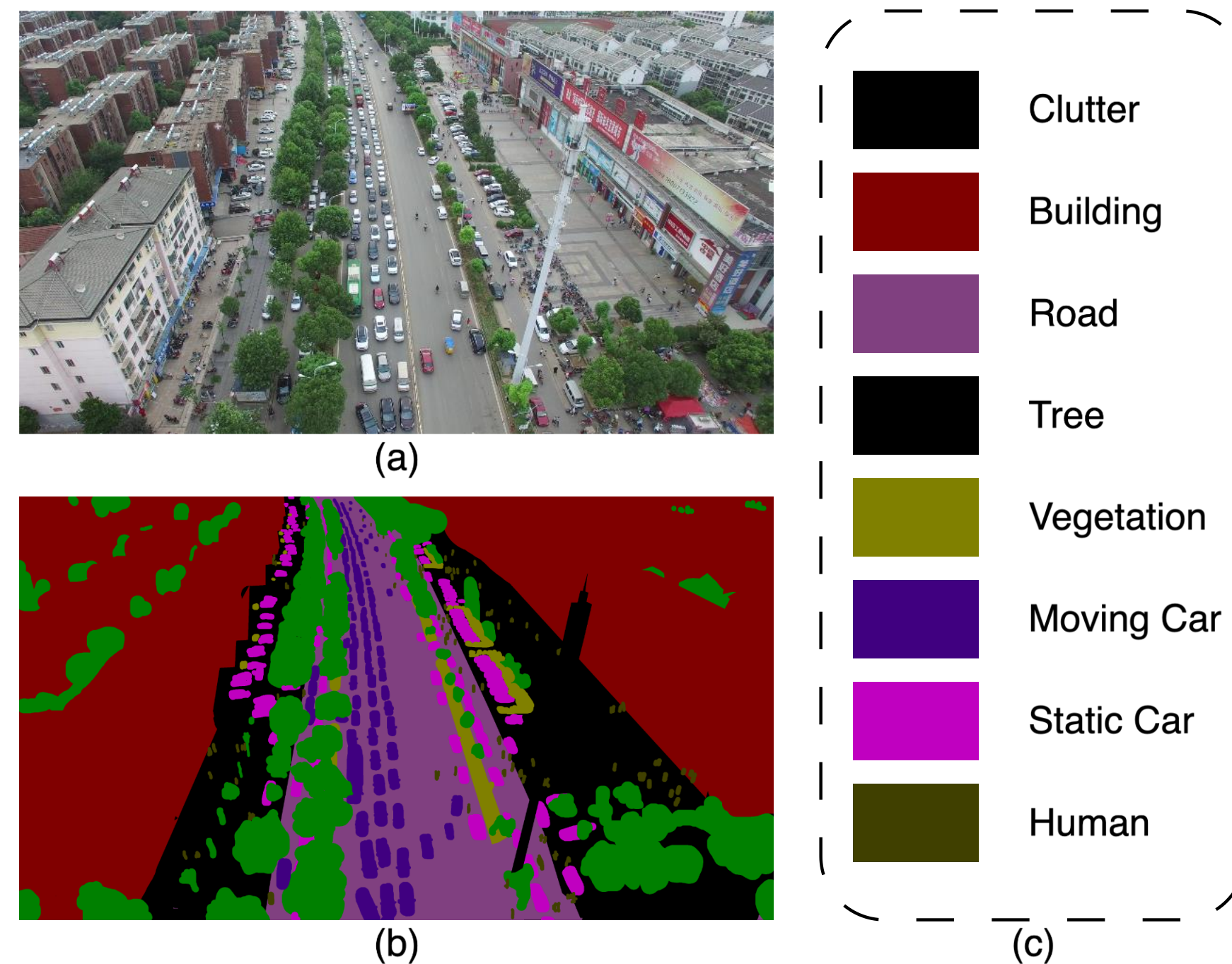


Figure 1. A sample of UAVid dataset. (a) the original image. (b) the semantic label. (c) color definition of semantic labels

Method

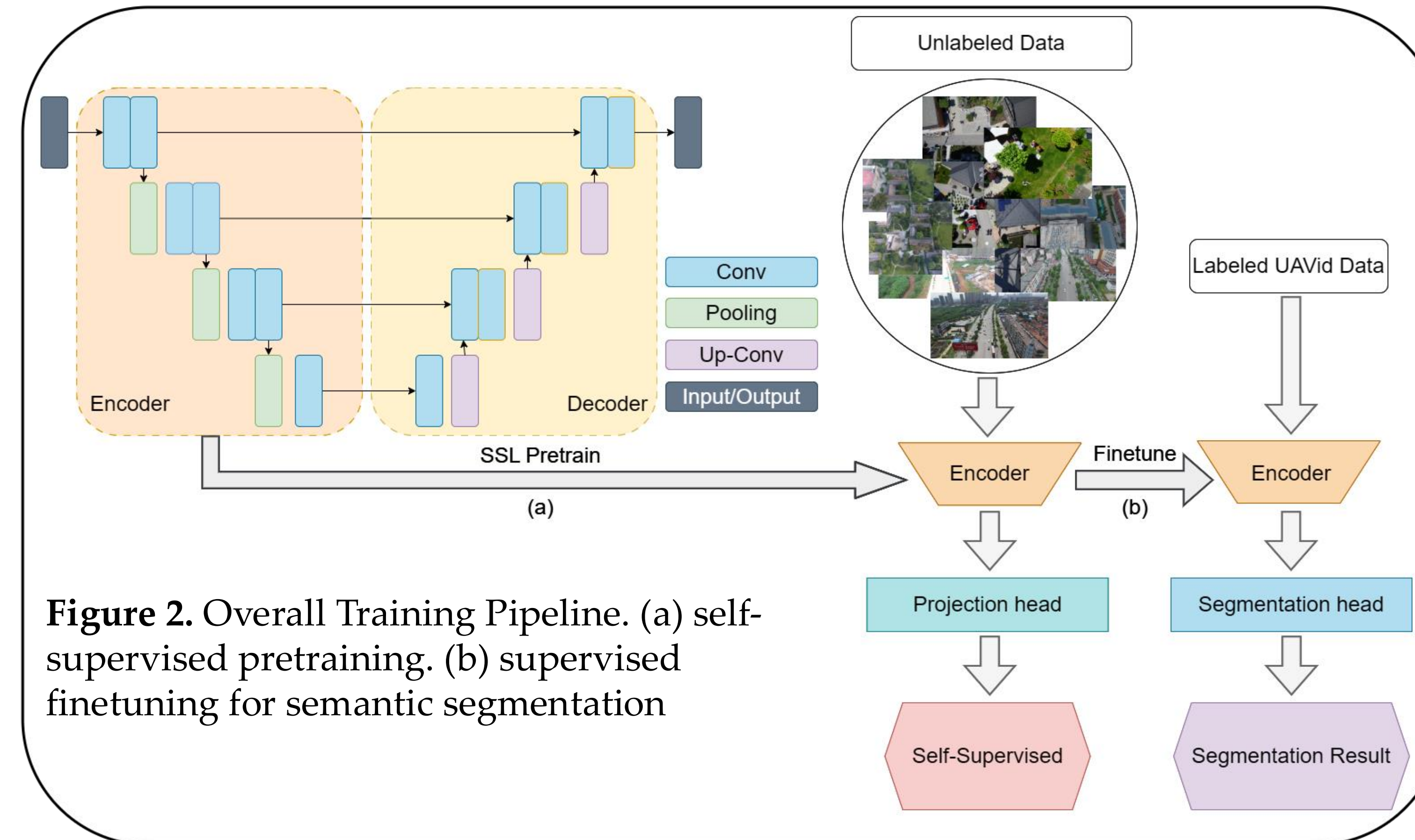


Figure 2. Overall Training Pipeline. (a) self-supervised pretraining. (b) supervised finetuning for semantic segmentation

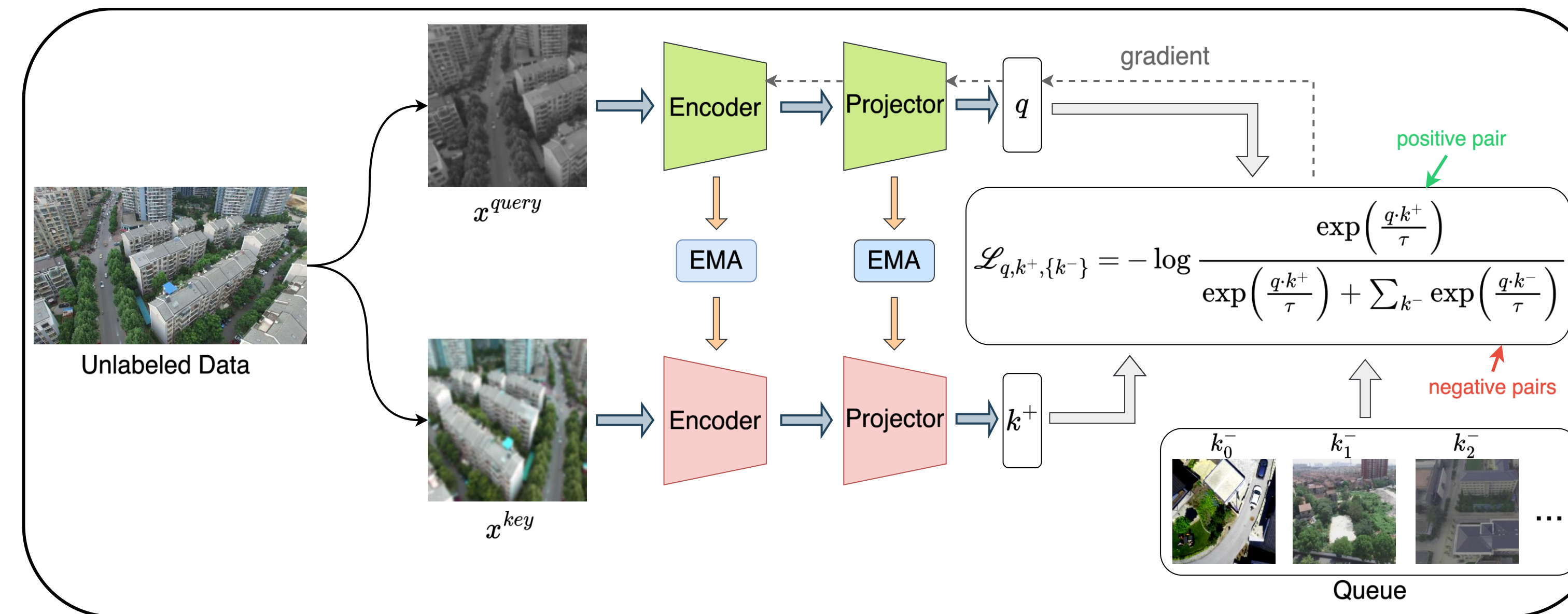


Figure 3. MoCo v2 Self-Supervised Pretraining Pipeline

❑ Exponential Moving Average:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$$

❑ Supervised Loss:

$$\mathcal{L}_{\text{sup}} = \lambda \mathcal{L}_{\text{Dice}} + (1 - \lambda) \mathcal{L}_{\text{CE}}$$

Experimental Results

Model	mIoU	F1	OA
DeepLabV3 + SSL	65.12	77.34	85.53
DeepLabV3 + ImageNet	64.35	76.70	85.06
UNet++ + SSL	63.99	76.64	84.49
UNet++ + ImageNet	60.28	74.84	82.00
UNet + SSL	64.15	76.72	84.94
UNet + ImageNet	61.75	74.72	83.22
UNet + RandomInit	41.69	62.97	69.96

Table 1. Summary of mean IoU, F1, and overall accuracy for all models

Model	Clutter	Building	Road	Tree	Vegetation	Moving Car	Static Car	Human
DeepLabV3 + SSL	66.02	86.49	78.84	78.83	62.69	70.49	48.82	28.78
DeepLabV3 + ImageNet	65.45	86.00	78.86	78.25	60.11	68.33	51.09	26.71
UNet++ + SSL	63.81	84.17	77.14	78.42	60.87	69.13	48.20	30.19
UNet++ + ImageNet	58.57	80.03	71.56	76.28	59.17	63.51	45.00	28.13
UNet + SSL	64.96	85.56	77.47	78.50	61.57	66.72	48.72	29.68
UNet + ImageNet	61.98	83.76	74.86	76.41	58.79	66.22	46.53	25.47
UNet + RandomInit	46.25	66.39	62.80	59.53	36.05	42.48	20.01	0.00

Table 2. Class-wise IoU comparison

❑ SSL Pretraining Setup (MoCo v2)

- Backbone: ResNet-50
- Batch size: 256
- Queue size: 4096
- Optimizer: SGD (LR 1e-3)

❑ Supervised Finetuning Setup

- Models: UNet, UNet++, DeepLabV3
- Patch size: 512x512
- Optimizer: AdamW

Experimental Results

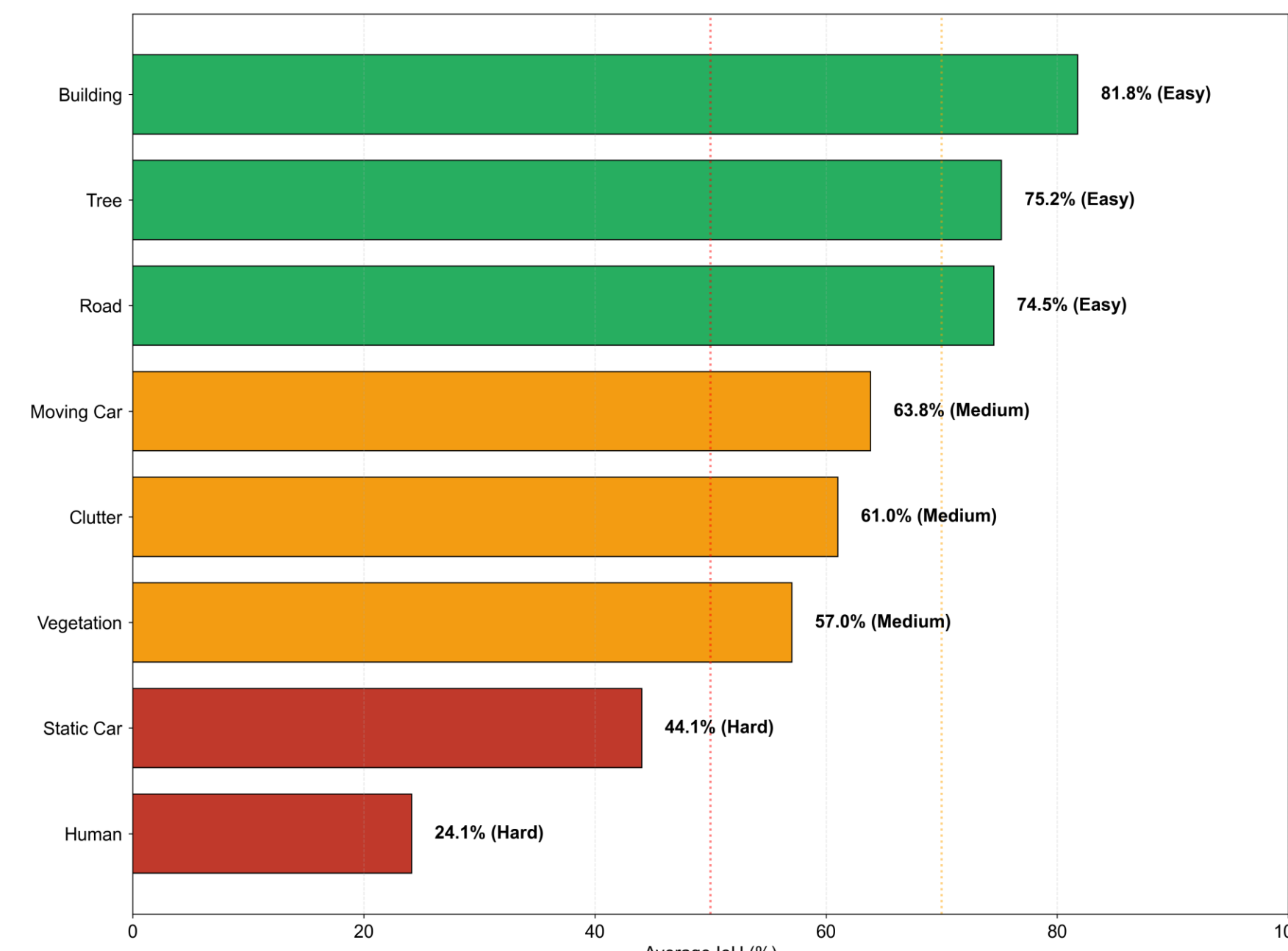


Figure 4. Segmentation Difficulty Across Classes



Figure 5. SSL vs ImageNet: Per-Class IoU Gains

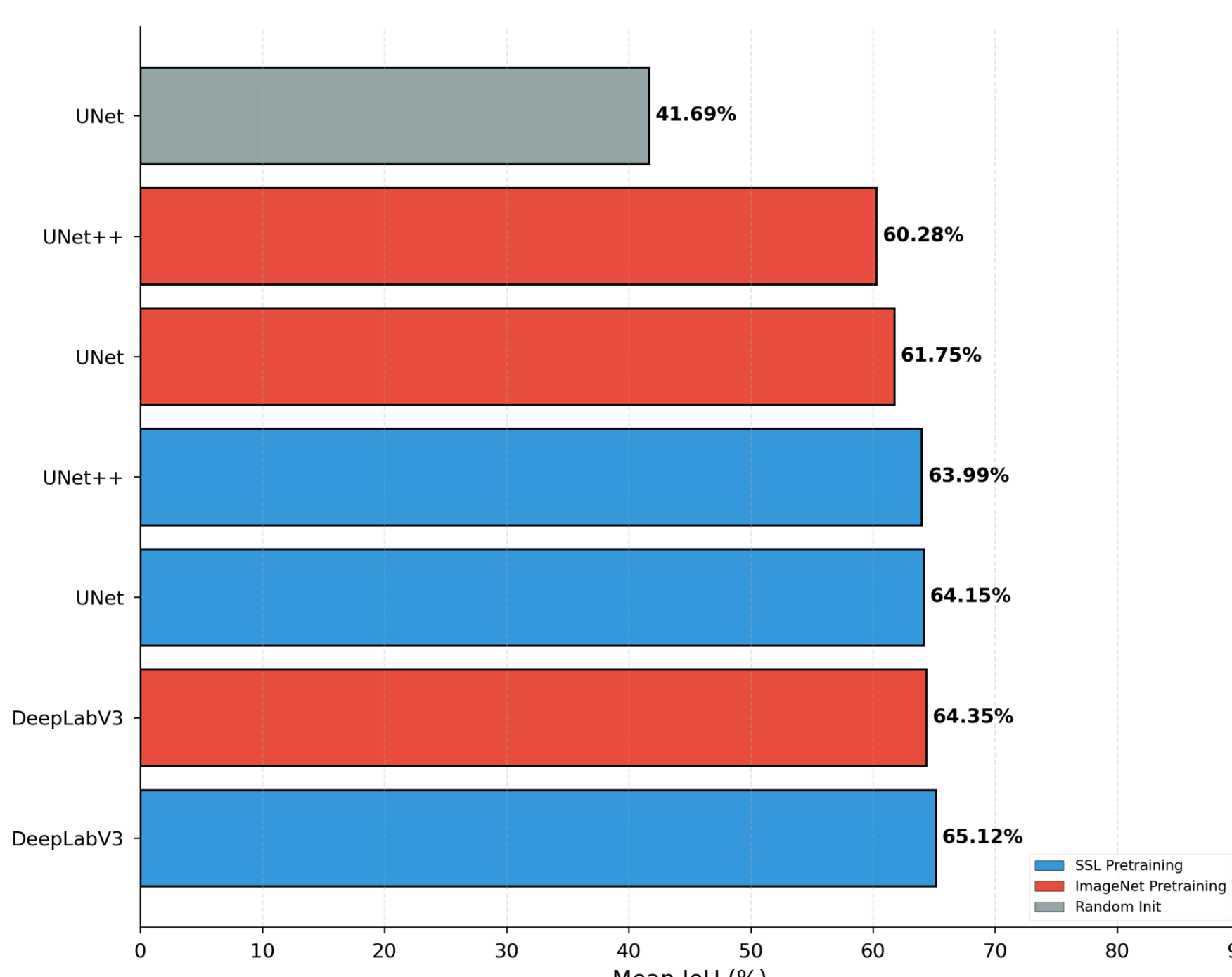


Figure 6. Comparison of Pretraining Strategies Across Model

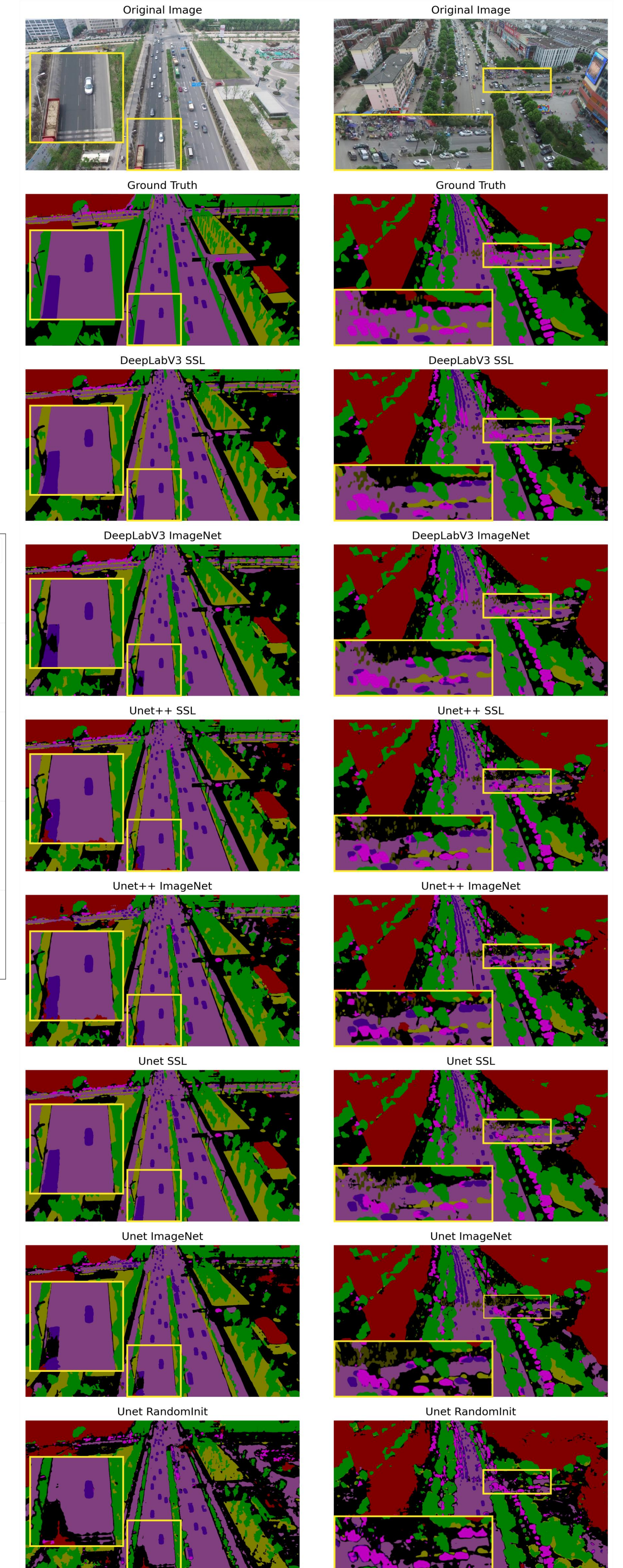


Figure 7. Visual Comparison

Future Work

- ❖ Scale unlabeled data and UAV-specific augmentations, incorporate stronger post-processing to further refine segmentation quality
- ❖ Adopt newer SSL methods
- ❖ Expand SSL pretraining to stronger architectures (e.g., SwinTransformer, SegFormer)
- ❖ Investigate UNet-2.5D to capture temporal continuity across consecutive UAV frames

